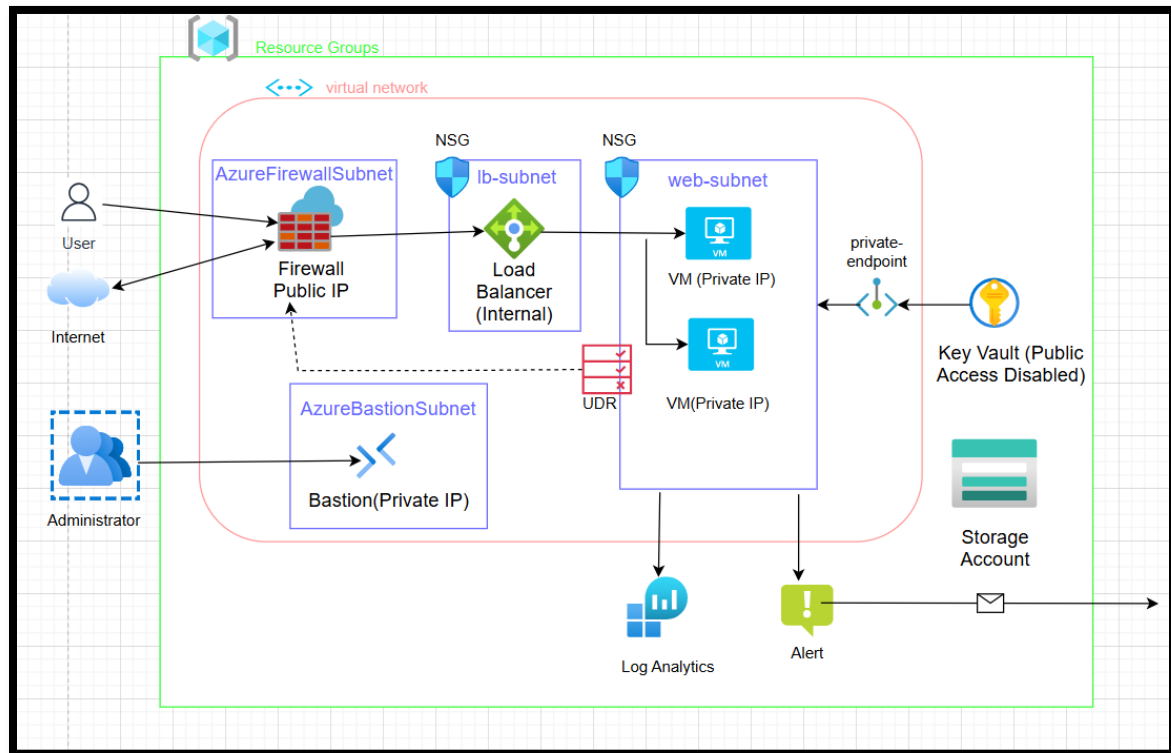


LOW LEVEL DESIGN (LLD)

1. Solution Overview

This infrastructure is a secure production-style Azure environment built using Terraform modular architecture.



The architecture includes:

- Azure Firewall for centralized traffic filtering
- Internal Load Balancer for backend traffic distribution
- Linux Virtual Machines inside private subnet
- Azure Bastion for secure administration
- Azure Key Vault with Private Endpoint integration
- Azure Storage Account
- Centralized monitoring using Log Analytics and Alerts
- NSG and UDR based traffic control

2. Resource Group

Resource	Value
----------	-------

Resource Group	dev-rg
----------------	--------

Region	East US
--------	---------

3. Virtual Network Design

Component	Address Space
-----------	---------------

Virtual Network	dev-vnet
-----------------	----------

CIDR	10.0.0.0/16
------	-------------

4. Subnet Design

Subnet Name	CIDR	Purpose
lb-subnet	10.0.1.0/24	Internal Load Balancer
web-subnet	10.0.2.0/24	Web Linux VMs
db-subnet	10.0.3.0/24	Database Layer
AzureBastionSubnet	10.0.4.0/27	Azure Bastion
AzureFirewallSubnet	10.0.5.0/24	Azure Firewall

5. Network Security Group (NSG) Design

lb-nsg

Applied To:

- lb-subnet

Rules

Rule Name	Priority	Port	Source	Action
-----------	----------	------	--------	--------

allow-http	100	80	Internet	Allow
------------	-----	----	----------	-------

allow-https	110	443	Internet	Allow
-------------	-----	-----	----------	-------

web-nsg

Applied To:

- web-subnet

Rules

Rule Name	Priority	Port	Source	Action
-----------	----------	------	--------	--------

web-http	100	80	AzureLoadBalancer	Allow
----------	-----	----	-------------------	-------

web-https	110	443	AzureLoadBalancer	Allow
-----------	-----	-----	-------------------	-------

allow-ssh	120	22	VirtualNetwork	Allow
-----------	-----	----	----------------	-------

6. Azure Firewall Design

Component	Value
-----------	-------

Firewall Name	dev-firewall
---------------	--------------

Public IP	firewall-pip
-----------	--------------

Subnet	AzureFirewallSubnet
--------	---------------------

Firewall Traffic Flow

Internet



Azure Firewall (Public IP)



Internal Load Balancer



Web Linux VMs

7. Load Balancer Design

Component	Value
Load Balancer Name	dev-lb
Type	Internal Load Balancer
Subnet	lb-subnet
SKU	Standard

Backend Pool

- web-vm-1
 - web-vm-2
-

Health Probe

Probe	Port
health-probe	80

8. Linux Virtual Machine Design

VM Name	Zone	Size
web-vm-1	Zone 1	Standard_B1s
web-vm-2	Zone 2	Standard_B1s

VM Security Controls

- No Public IP
- Private Subnet Deployment
- Bastion-based administration

- Internal LB traffic only

9. Azure Bastion Design

Component Value

Bastion Name dev-bastion

Subnet AzureBastionSubnet

Public IP bastion-pip

Administrative Access Flow

Administrator

↓

Azure Bastion

↓

Linux VMs

10. Azure Key Vault Design

Component Value

Key Vault dev-keyvault-mahi

Access Private Endpoint

Stored Secrets

Secret

vm-username

vm-password

11. Private Endpoint Design

Component	Value
Private Endpoint	kv-private-endpoint
DNS Zone	privatelink.vaultcore.azure.net

Private Connectivity Flow

Linux VM
↓
Private Endpoint
↓
Azure Key Vault

12. Storage Account Design

Component	Value
Storage Account	mahistoragedev01
Tier	Standard
Replication	LRS
Kind	StorageV2

Storage Access

Current design uses:

- Public Endpoint access
-

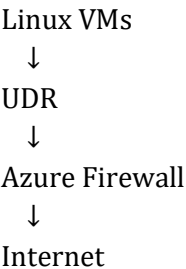
13. User Defined Route (UDR) Design

Component	Value
Route Table	web-udr
Applied To	web-subnet

Route Configuration

Prefix	Next Hop
0.0.0.0/0	Azure Firewall

Outbound Traffic Flow



14. Monitoring and Alerting Design

Log Analytics

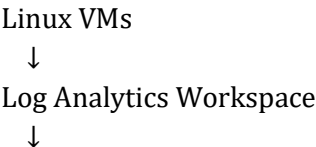
Workspace

dev-log-workspace

Alerts

Alert	VM
high-cpu-alert-vm1	web-vm-1
high-cpu-alert-vm2	web-vm-2

Monitoring Flow

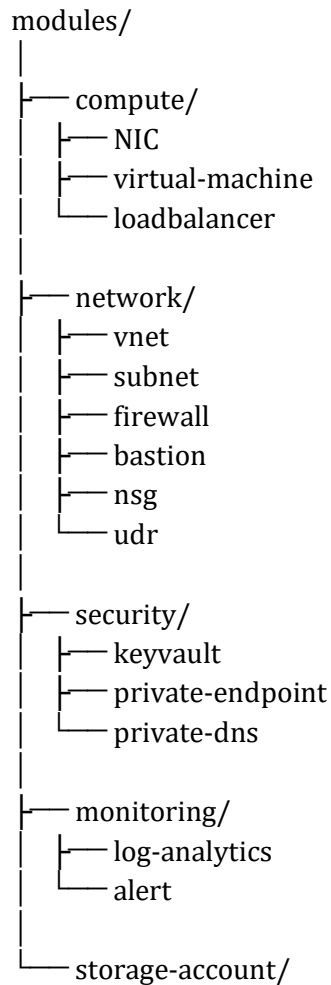


Azure Monitor Alerts



Email Notification

15. Terraform Modular Architecture



16. Security Best Practices Implemented

- Private VM deployment
- Bastion-only administration
- Firewall-controlled ingress and egress traffic
- Internal Load Balancer architecture
- NSG-based subnet protection

- Private Endpoint for Key Vault
 - Centralized monitoring and alerting
-

17. Final Traffic Summary

Inbound Flow

User
↓
Internet
↓
Azure Firewall
↓
Internal Load Balancer
↓
Linux VMs

Admin Flow

Administrator
↓
Azure Bastion
↓
Linux VMs

Secret Flow

Linux VMs
↓
Private Endpoint
↓
Azure Key Vault